

Notebook 03: Reproducible Incomplete-Lead Simulation Engine

Project: Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms

1. Overview & Research Objectives

In real-world emergency, outpatient, and ambulatory cardiac monitoring, full 12-lead ECGs are frequently incomplete due to detached electrodes, motion artifacts, emergency triage constraints, or wearable device limitations (e.g. single-lead smartwatches or 3-lead patches).

A truly **lead-agnostic** cardiac AI must reliably infer multi-label pathology across:

1. **Complete 12-Lead ECG** (12/12 leads available).
2. **Systematic Lead Reductions**: Evaluation across $k \in \{11, 10, 9, 8, 6, 4, 3, 2, 1\}$ available leads.
3. **Leave-One-Out Sensitivity**: Isolating the marginal diagnostic contribution of each individual lead ($L \in \{I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6\}$).
4. **Clinically Structured Subsets**:
 - **Limb Leads Only** (Leads I, II, III, aVR, aVL, aVF) - representative of standard limb cable setups.
 - **Precordial Leads Only** (Leads V1 - V6) - representative of anterior/septal/lateral chest electrode arrays.
 - **Einthoven Triangle** (Leads I, II, III) - standard 3-lead peripheral cardiac monitoring.
 - **Rhythm Lead II Only** - standard single-channel telemetry strip.
 - **Wearable Patch / Smartwatch (Lead I Only)** - single-channel bilateral arm configuration.

In this notebook, we design a mathematically formal, reproducible masking protocol: $X_{\text{masked}} = X \odot M, \quad M \in \{0, 1\}^{B \times 12}$ and export benchmark masks to `results/metrics/incomplete_lead_masks.npz` and scenario definitions to `results/tables/incomplete_lead_scenarios.csv`.

```
In [1]: import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('seaborn-v0_8-whitegrid' if 'seaborn-v0_8-whitegrid' in plt.style.ava
```

```
plt.rcParams.update({
    'font.sans-serif': 'DejaVu Sans',
    'font.size': 11,
    'axes.labelsize': 12,
    'axes.titlesize': 13,
    'xtick.labelsize': 10,
    'ytick.labelsize': 10,
    'figure.titlesize': 14,
    'figure.dpi': 300
})

RESULTS_DIR = r"C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Mo
FIG_DIR = os.path.join(REULTS_DIR, "figures")
TAB_DIR = os.path.join(REULTS_DIR, "tables")
MET_DIR = os.path.join(REULTS_DIR, "metrics")

print("Incomplete-lead engine initialized.")
```

Incomplete-lead engine initialized.

```
In [2]: LEAD_NAMES = ['I', 'II', 'III', 'aVR', 'aVL', 'aVF', 'V1', 'V2', 'V3', 'V4', 'V5',
LEAD_INDEX = {name: i for i, name in enumerate(LEAD_NAMES)}

# Define standard clinical subsets
CLINICAL_SCENARIOS = {
    'Full_12_Lead': LEAD_NAMES,
    'Limb_Only_6_Lead': ['I', 'II', 'III', 'aVR', 'aVL', 'aVF'],
    'Precordial_Only_6_Lead': ['V1', 'V2', 'V3', 'V4', 'V5', 'V6'],
    'Einthoven_3_Lead': ['I', 'II', 'III'],
    'Lead_II_Rhythm_Strip': ['II'],
    'Wearable_Lead_I': ['I'],
    'Contiguous_Anterior_V1_V3': ['V1', 'V2', 'V3'],
    'Contiguous_Lateral_I_aVL_V5_V6': ['I', 'aVL', 'V5', 'V6'],
    'Contiguous_Inferior_II_III_aVF': ['II', 'III', 'aVF']
}

print("Standard clinical ECG lead groups defined:")
for sc_name, leads in CLINICAL_SCENARIOS.items():
    print(f" {sc_name:30s}: {len(leads):2d} leads -> {' '.join(leads)}")
```

Standard clinical ECG lead groups defined:

```
Full_12_Lead          : 12 leads -> I, II, III, aVR, aVL, aVF, V1, V2, V3,
V4, V5, V6
Limb_Only_6_Lead      : 6 leads -> I, II, III, aVR, aVL, aVF
Precordial_Only_6_Lead : 6 leads -> V1, V2, V3, V4, V5, V6
Einthoven_3_Lead      : 3 leads -> I, II, III
Lead_II_Rhythm_Strip   : 1 leads -> II
Wearable_Lead_I        : 1 leads -> I
Contiguous_Anterior_V1_V3 : 3 leads -> V1, V2, V3
Contiguous_Lateral_I_aVL_V5_V6: 4 leads -> I, aVL, V5, V6
Contiguous_Inferior_II_III_aVF: 3 leads -> II, III, aVF
```

```
In [3]: # Systematic missing Lead levels: k available Leads
k_levels = [12, 11, 10, 9, 8, 6, 4, 3, 2, 1]

# Load test set size to create reproducible scenario masks
npz_path = os.path.join(MET_DIR, "preprocessed_data.npz")
```

```

data = np.load(npz_path)
N_test = len(data['y_test'])
print(f"Generating reproducible scenario masks for N_test = {N_test:,} samples.")

np.random.seed(42)
scenario_masks = {}

# 1. Full 12 Leads
scenario_masks['full_12'] = np.ones((N_test, 12), dtype=np.float32)

# 2. Fixed Clinical Groups
for sc_name, leads in CLINICAL_SCENARIOS.items():
    mask = np.zeros((N_test, 12), dtype=np.float32)
    indices = [LEAD_INDEX[l] for l in leads]
    mask[:, indices] = 1.0
    scenario_masks[sc_name] = mask

# 3. Systematic k-Lead missing (randomly chosen per sample with seed 42)
for k in k_levels:
    if k == 12: continue
    mask = np.zeros((N_test, 12), dtype=np.float32)
    for i in range(N_test):
        avail_indices = np.random.choice(12, k, replace=False)
        mask[i, avail_indices] = 1.0
    scenario_masks[f'random_{k}_leads'] = mask

# 4. Leave-one-out individual lead sensitivity (11 Leads available, 1 Lead masked)
for lead in LEAD_NAMES:
    mask = np.ones((N_test, 12), dtype=np.float32)
    mask[:, LEAD_INDEX[lead]] = 0.0
    scenario_masks[f'drop_lead_{lead}'] = mask

print(f"Total evaluated missing-lead scenarios generated: {len(scenario_masks)}")

```

Generating reproducible scenario masks for N_test = 525 samples.
Total evaluated missing-lead scenarios generated: 31

```

In [4]: scenario_summary = []
for name, mask in scenario_masks.items():
    mean_leads = mask.sum(axis=1).mean()
    if name in CLINICAL_SCENARIOS:
        cat = 'Clinical Standard Group'
        desc = f"Fixed leads: {'', '.join(CLINICAL_SCENARIOS[name])}"
    elif name.startswith('random_'):
        cat = 'Systematic Random Reduction'
        desc = f"Random {int(mean_leads)} leads available per sample"
    elif name.startswith('drop_'):
        cat = 'Leave-One-Out Sensitivity'
        desc = f"All leads present except {name.split('drop_lead_')[1]}"
    else:
        cat = 'Reference Complete ECG'
        desc = "Standard full 12 leads"

    scenario_summary.append({
        'Scenario Identifier': name,
        'Category': cat,

```

```
        'Mean Leads Available': round(mean_leads, 1),
        'Missing Leads': 12 - int(mean_leads),
        'Description': desc
    })

df_scenarios = pd.DataFrame(scenario_summary)
out_csv = os.path.join(TAB_DIR, "incomplete_lead_scenarios.csv")
df_scenarios.to_csv(out_csv, index=False)
print("=== INCOMPLETE LEAD SCENARIO DEFINITIONS ===")
print(df_scenarios.to_string(index=False))
print(f"\nSaved scenario definitions to: {out_csv}")
```

=== INCOMPLETE LEAD SCENARIO DEFINITIONS ===

Scenario Identifier	Category	Mean Leads Available	Mi
ssing Leads	Description		
full_12	Reference Complete ECG	12.0	
0	Standard full 12 leads		
Full_12_Lead	Clinical Standard Group	12.0	
0	Fixed leads: I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, V6		
Limb_Only_6_Lead	Clinical Standard Group	6.0	
6	Fixed leads: I, II, III, aVR, aVL, aVF		
Precordial_Only_6_Lead	Clinical Standard Group	6.0	
6	Fixed leads: V1, V2, V3, V4, V5, V6		
Einthoven_3_Lead	Clinical Standard Group	3.0	
9	Fixed leads: I, II, III		
Lead_II_Rhythm_Strip	Clinical Standard Group	1.0	
11	Fixed leads: II		
Wearable_Lead_I	Clinical Standard Group	1.0	
11	Fixed leads: I		
Contiguous_Anterior_V1_V3	Clinical Standard Group	3.0	
9	Fixed leads: V1, V2, V3		
Contiguous_Lateral_I_aVL_V5_V6	Clinical Standard Group	4.0	
8	Fixed leads: I, aVL, V5, V6		
Contiguous_Inferior_II_III_aVF	Clinical Standard Group	3.0	
9	Fixed leads: II, III, aVF		
random_11_leads	Systematic Random Reduction	11.0	
1	Random 11 leads available per sample		
random_10_leads	Systematic Random Reduction	10.0	
2	Random 10 leads available per sample		
random_9_leads	Systematic Random Reduction	9.0	
3	Random 9 leads available per sample		
random_8_leads	Systematic Random Reduction	8.0	
4	Random 8 leads available per sample		
random_6_leads	Systematic Random Reduction	6.0	
6	Random 6 leads available per sample		
random_4_leads	Systematic Random Reduction	4.0	
8	Random 4 leads available per sample		
random_3_leads	Systematic Random Reduction	3.0	
9	Random 3 leads available per sample		
random_2_leads	Systematic Random Reduction	2.0	
10	Random 2 leads available per sample		
random_1_leads	Systematic Random Reduction	1.0	
11	Random 1 leads available per sample		
drop_lead_I	Leave-One-Out Sensitivity	11.0	
1	All leads present except I		
drop_lead_II	Leave-One-Out Sensitivity	11.0	
1	All leads present except II		
drop_lead_III	Leave-One-Out Sensitivity	11.0	
1	All leads present except III		
drop_lead_aVR	Leave-One-Out Sensitivity	11.0	
1	All leads present except aVR		
drop_lead_aVL	Leave-One-Out Sensitivity	11.0	
1	All leads present except aVL		
drop_lead_aVF	Leave-One-Out Sensitivity	11.0	
1	All leads present except aVF		
drop_lead_V1	Leave-One-Out Sensitivity	11.0	
1	All leads present except V1		
drop_lead_V2	Leave-One-Out Sensitivity	11.0	

1		All leads present except V2	
	drop_lead_V3	Leave-One-Out Sensitivity	11.0
1		All leads present except V3	
	drop_lead_V4	Leave-One-Out Sensitivity	11.0
1		All leads present except V4	
	drop_lead_V5	Leave-One-Out Sensitivity	11.0
1		All leads present except V5	
	drop_lead_V6	Leave-One-Out Sensitivity	11.0
1		All leads present except V6	

Saved scenario definitions to: C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms\results\tables\incomplete_lead_scenarios.csv

```
In [5]: # Visualize sample ECG under various missing lead masks
X_sample = data['X_test'][0] # (12, 1000)
t_sec = np.arange(500) / 100.0 # First 5 seconds

fig, axes = plt.subplots(6, 1, figsize=(16, 14), sharex=True)

scenarios_to_plot = [
    ('Complete 12-Lead ECG (12/12 Available)', np.ones(12)),
    ('Limb Leads Only (I, II, III, aVR, aVL, aVF)', scenario_masks['Limb_Only_6_Leads'][0]),
    ('Precordial Leads Only (V1 - V6)', scenario_masks['Precordial_Only_6_Lead'][0]),
    ('Einthoven 3-Lead Monitoring (I, II, III)', scenario_masks['Einthoven_3_Lead'][0]),
    ('Systematic Random 4 Leads Available', scenario_masks['random_4_leads'][0]),
    ('Wearable / Telemetry (Lead II Only)', scenario_masks['Lead_II_Rhythm_Strip'][0])
]

lead_colors = sns.color_palette("tab20", 12)

for ax_idx, (title, mask_vec) in enumerate(scenarios_to_plot):
    ax = axes[ax_idx]
    active_leads = [LEAD_NAMES[i] for i in range(12) if mask_vec[i] == 1.0]

    # Plot active leads with vertical offset
    offset = 0
    for l_idx in range(12):
        if mask_vec[l_idx] == 1.0:
            sig = X_sample[l_idx, :500] + offset
            ax.plot(t_sec, sig, color=lead_colors[l_idx], linewidth=1.1, label=f"Lead {l_idx}")
            offset += 3.0

    ax.set_title(f"{title} | Active: [{', '.join(active_leads)}]", fontweight='bold')
    ax.set_ylabel('Amplitude (a.u.)')
    ax.set_yticks([])
    ax.grid(True, linestyle=':', alpha=0.5)
    ax.legend(loc='upper right', ncol=min(6, len(active_leads)), fontsize=8, frameon=False)

axes[-1].set_xlabel('Time (seconds)')
plt.suptitle('Controlled Incomplete 12-Lead ECG Missingness Configurations', fontsize=12)
plt.tight_layout()
fig5_path = os.path.join(FIG_DIR, "fig5_incomplete_lead_scenarios.png")
plt.savefig(fig5_path, dpi=300, bbox_inches='tight')
plt.show()
print(f"Figure 5 saved to: {fig5_path}")
```

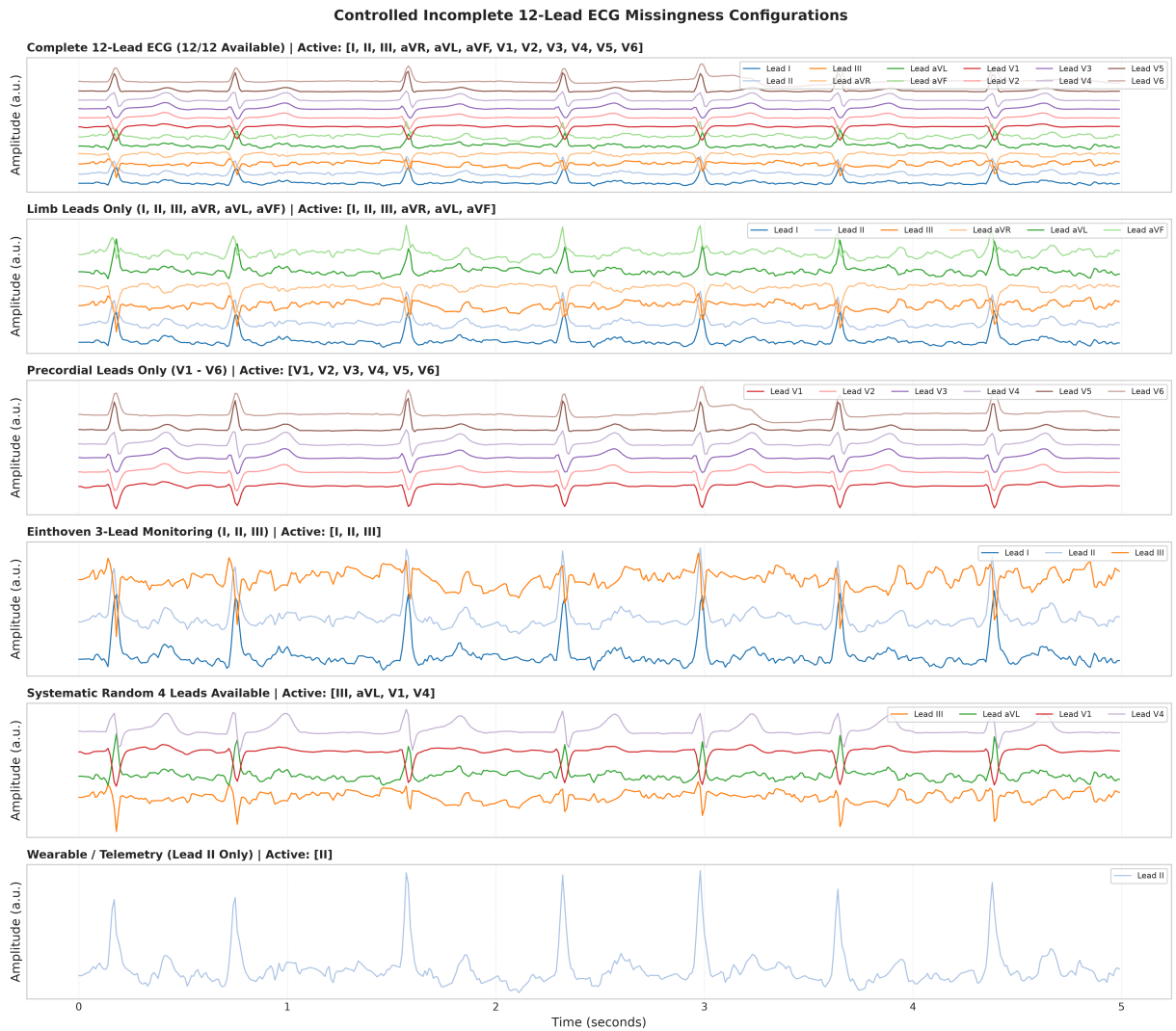


Figure 5 saved to: C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms\results\figures\fig5_incomplete_lead_scenarios.png

```
In [6]: out_masks_npz = os.path.join(MET_DIR, "incomplete_lead_masks.npz")
np.savez_compressed(out_masks_npz, **scenario_masks)
print(f"Successfully serialized {len(scenario_masks)} test scenario masks to: {out_masks_npz}")
print(f"File size: {os.path.getsize(out_masks_npz)/1024:.2f} KB")
```

Successfully serialized 31 test scenario masks to: C:\Research\Lead-Agnostic and Uncertainty-Calibrated State-Space Modeling for Multi-Label Cardiac Abnormality Detection from Incomplete 12-Lead Electrocardiograms\results\metrics\incomplete_lead_masks.npz
File size: 19.28 KB